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# Classification of the three flower types in Iris Dataset using SVM

# Import libraries
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

# Load Iris dataset
iris = datasets.load_iris()

# Features (X) and Target (y)
X = iris.data
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create SVM model
model = SVC(kernel='linear')

# Train model
model.fit(X_train, y_train)

# Predict on test data
y_pred = model.predict(X_test)

# Evaluate model
print("Accuracy:", accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Predict a new sample
sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(sample)

print("\nPredicted Class:", iris.target_names[prediction][0])

# Classification of the Three Flower Types in Iris Dataset using Naive
Bayes

# Step 1: Import Libraries
from sklearn.datasets import load_iris

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from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

# Step 2: Load Iris Dataset
iris = load_iris()

X = iris.data      # Features
y = iris.target    # Target classes

# Feature names:
print("Features:", iris.feature_names)

# Class names:
print("Classes:", iris.target_names)

# Step 3: Split Dataset into Training and Testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=42
)

# Step 4: Create Naive Bayes Model
# GaussianNB is used because Iris features are continuous values
model = GaussianNB()

# Step 5: Train the Model
model.fit(X_train, y_train)

# Step 6: Make Predictions
y_pred = model.predict(X_test)

# Step 7: Evaluate the Model
accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Classification of the three flower types in Iris Dataset
# using Logistic Regression

# Step 1: Import libraries

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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

# Step 2: Load Iris dataset
iris = load_iris()

X = iris.data          # Features
y = iris.target        # Target (3 flower types)

# Step 3: Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Step 4: Create Logistic Regression model
model = LogisticRegression(max_iter=200)

# Step 5: Train the model
model.fit(X_train, y_train)

# Step 6: Predict on test data
y_pred = model.predict(X_test)

# Step 7: Evaluate the model
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix:\n")
print(confusion_matrix(y_test, y_pred))

# Step 8: Predict a new sample
sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(sample)

print("\nPredicted Class:", iris.target_names[prediction[0]])

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